

# **RAI-818: INTELLIGENT** **MOBILE ROBOTICS**

## **Probabilistic Robotics – I**

- **Localization**
- **Probability**
- **Bayesian Filtering**

**Text : Principles of Robot Motion – Theory, Algorithms and Implementations**  
**H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.**

***INSTRUCTOR:***

***DR. KUNWAR FARAZ AHMED KHAN***

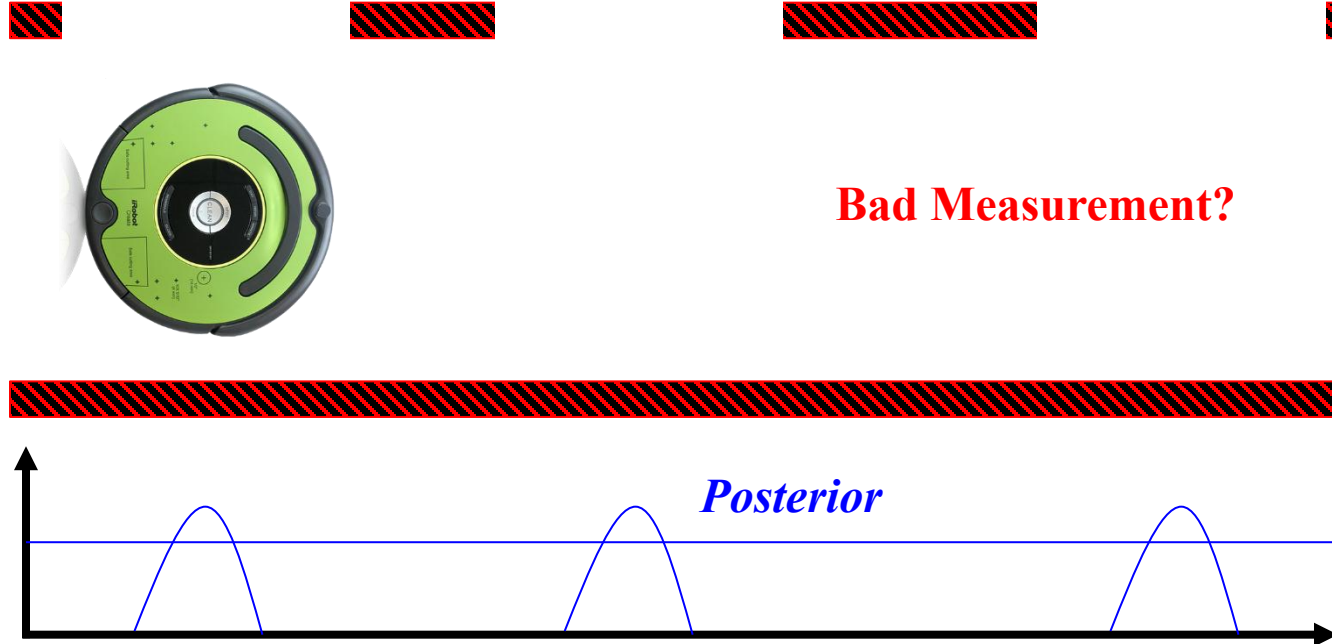
# Localization



- Localization
  - Basic Problem

## The Basic problem

- *Localization* is the ability of a machine to locate itself in space.
- Consider a robot lost in space. Within its environment, how can the robot locate where it is in space?
- *Uniform probability function*
- *Posterior belief*



Defining futures

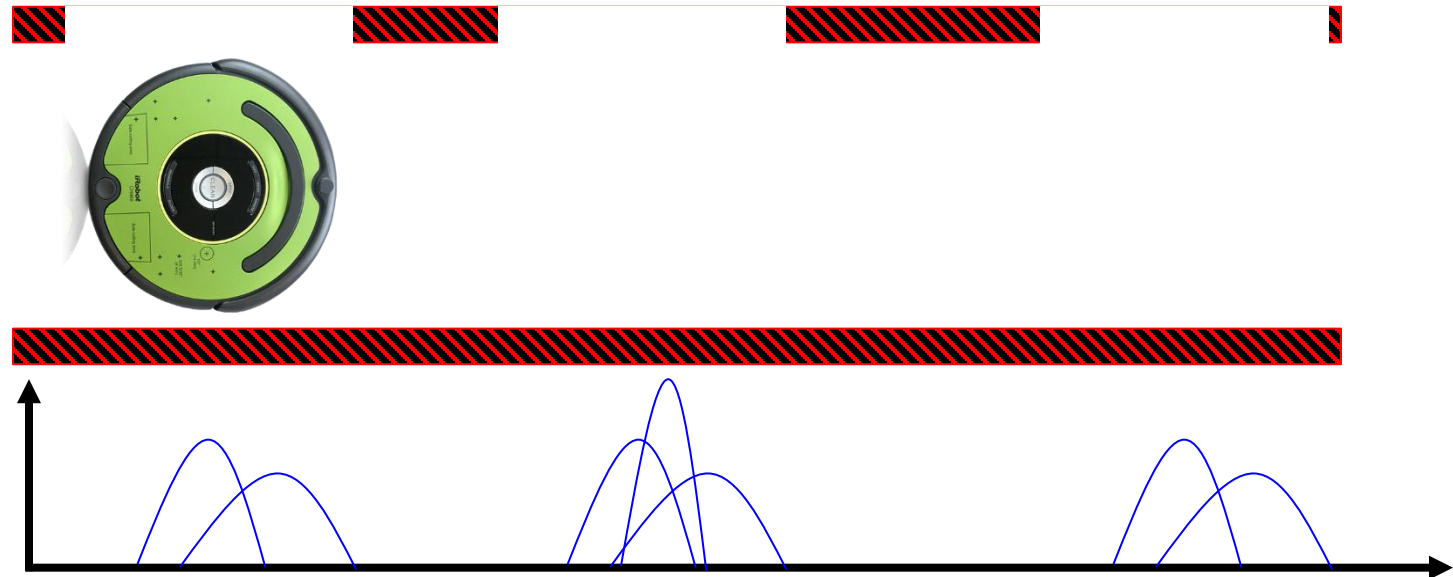
# Localization



- Localization
  - Basic Problem
  - **Robot Movement**

## Robot Movement

- Consider that the robot moves right a certain distance, we can shift the belief according to the motion.
- When we shift and flatten these bumps, we are performing convolution
- Now, assume that after the robot moves it senses itself right next to a door again so that the measurement is the same as before



Defining futures

# Localization



- Localization
  - Basic Problem
  - Robot Movement

## Robot Movement

- Now that we can establish the robot's initial belief about its location in the world, we want to update these beliefs given a measurement from the robot's sensors.
- Examine the measurement of the robot in a world with five cells,  $x_1$  through  $x_5$ .
- Say we know that externally two of the cells ( $x_2$  and  $x_3$ ) are colored red, and the other three ( $x_1$ ,  $x_4$  and  $x_5$ ) are green. We also know that our robot senses that it resides in a red cell.
- How will this affect the robot's belief in its location in the world?
- Knowing that the robot senses itself in a red cell, let's update the belief vector such that we are more likely to reside in a red cell and less likely to reside in a green cell.





# Localization

- Localization
  - Basic Problem
  - Robot Movement

## Robot Movement

- Lets come up with a simple rule to represent the probability that the robot is in a red or a green cell, based on the robot's measurement of 'red':

**red cells \* 0.6**

**green cells \* 0.2**

- For now, we have chosen these somewhat arbitrarily, but since 0.6 is three times larger than 0.2, this should increase the probability of being in a red cell by something like a factor of 3.
- Keep in mind that it is possible that our sensors are incorrect, so we do not want to multiply the green cell probability by zero.
- What is the probability that the robot is in a red cell and the probability that it's in a green cell?

**For red cells,  $0.2 * 0.6 = .12$**

**For green cells  $0.2 * 0.2 = 0.04$**

**So are these our probabilities?**

|       |       |       |       |       |
|-------|-------|-------|-------|-------|
| 0.04  | 0.12  | 0.12  | 0.04  | 0.04  |
| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ |



Defining futures

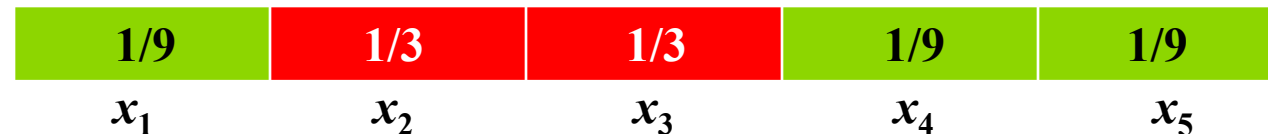
# Localization



- Localization
  - Basic Problem
  - **Robot Movement**

## Robot Movement

- No! The actual probabilities are  
**red cells = 1/3**      **green cells = 1/9**
- This is a probability distribution written as:  $P(X_i | Z)$ ,
- And read as The *posterior distribution* of place  $X_i$  given measurement  $Z$ ,  $Z$  being the measurement from the robot's sensors.
- This is an example of conditional probability.



Defining futures

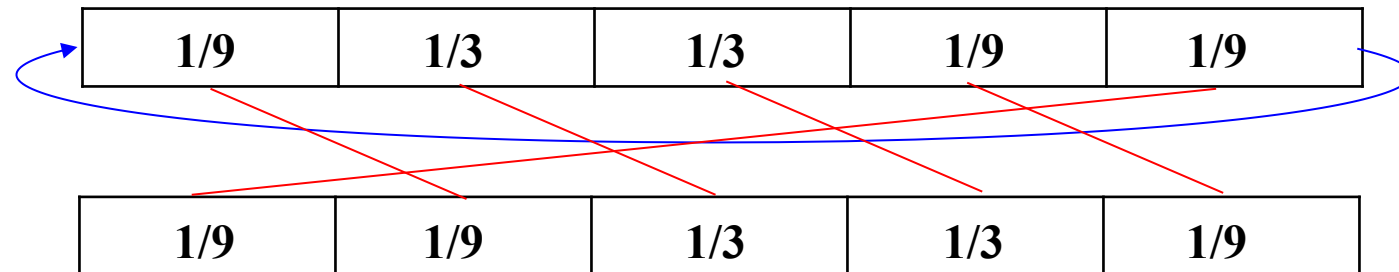
# Localization



- Localization
  - Basic Problem
  - **Robot Movement**

## Robot Movement

- Suppose there is a distribution of the cells as shown
- Assume the world is cyclic. So, when the robot reaches the right-most cell, it will circle around back to the first, or left-most cell.
- Assuming the robot moves exactly one position to the right, what is the posterior probability distribution after the motion?



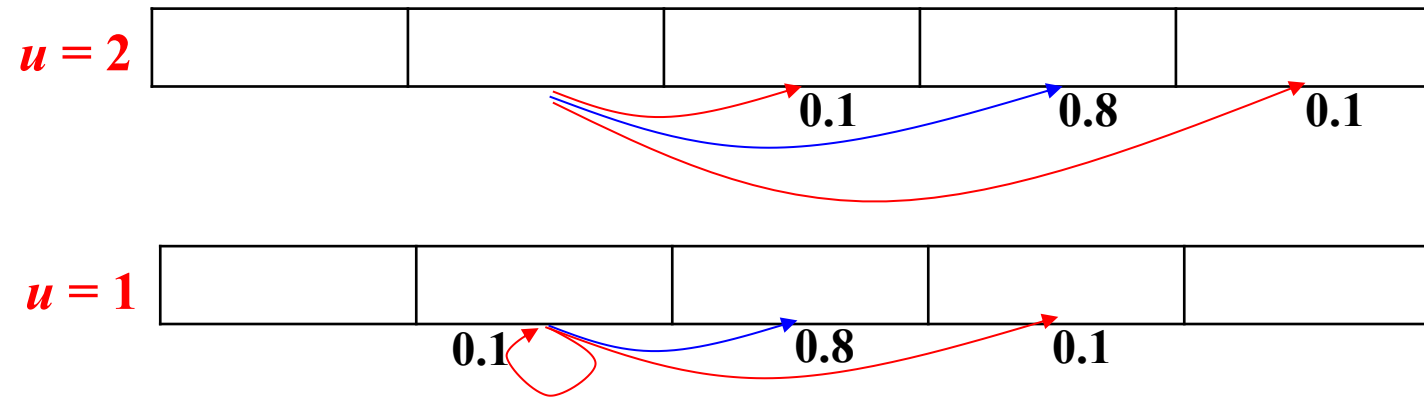
# Localization



- Localization
  - Basic Problem
  - **Robot Movement**

## Robot Movement

- Assume the robot executes its action with high probability (0.8), correctly; with small probability (0.1) it will overshoot or undershoot. This is a more accurate model of real robot motion.



- Given a prior distribution, can you give the distribution after the motion?



Defining futures





# Localization

- Localization
  - Basic Problem
  - **Robot Movement**

## Robot Movement

- Given a prior distribution, can you give the distribution after the motion?

**$u = 2$**

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 0 | 0 | 0 |
|---|---|---|---|---|

$$P(X_{i+2} | X_i) = 0.8$$

$$P(X_{i+1} | X_i) = 0.1$$

$$P(X_{i+3} | X_i) = 0.1$$

|   |   |     |     |     |
|---|---|-----|-----|-----|
| 0 | 0 | 0.1 | 0.8 | 0.1 |
|---|---|-----|-----|-----|

**$u = 2$**

|   |     |   |     |   |
|---|-----|---|-----|---|
| 0 | 0.5 | 0 | 0.5 | 0 |
|---|-----|---|-----|---|

$$P(X_{i+2} | X_i) = 0.8$$

$$P(X_{i+1} | X_i) = 0.1$$

$$P(X_{i+3} | X_i) = 0.1$$

|     |      |      |     |     |
|-----|------|------|-----|-----|
| 0.4 | 0.05 | 0.05 | 0.4 | 0.1 |
|-----|------|------|-----|-----|



Defining futures



# Localization

- Localization
  - Basic Problem
  - **Robot Movement**

## Robot Movement

- Given a prior distribution, can you give the distribution after the motion?

**$u = 2$**

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
|-----|-----|-----|-----|-----|

$$P(X_{i+2} | X_i) = 0.8$$

$$P(X_{i+1} | X_i) = 0.1$$

$$P(X_{i+3} | X_i) = 0.1$$

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
|-----|-----|-----|-----|-----|



Defining futures

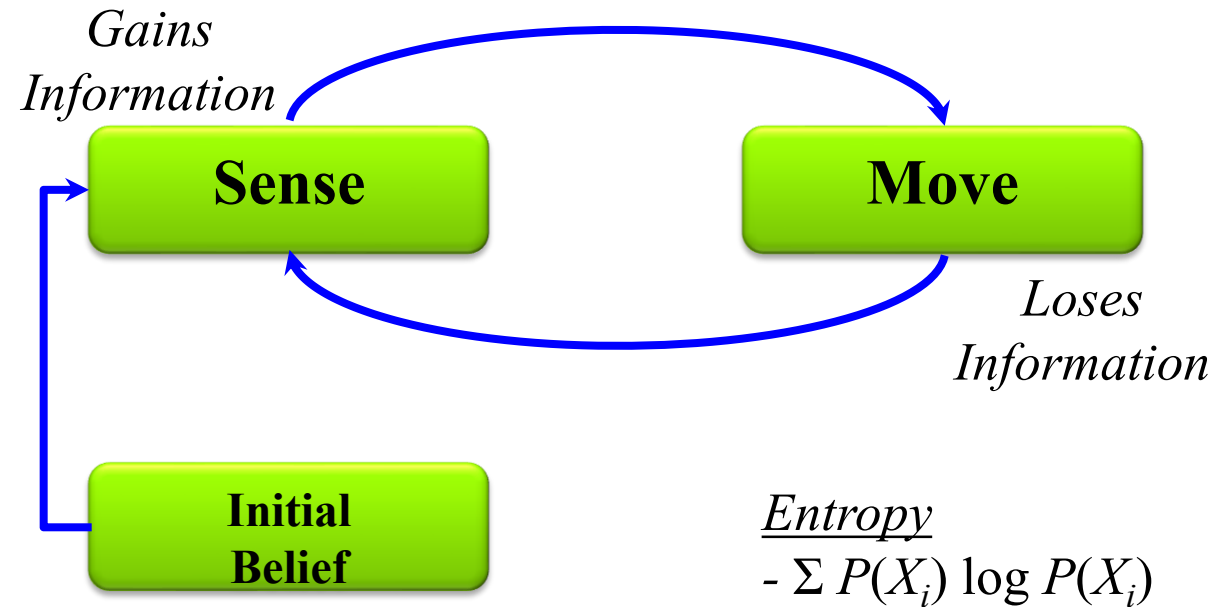
# Localization



- Localization
  - Basic Problem
  - Robot Movement
  - **Robot Sense and Movement**

## Robot Sense and Movement

- Monte Carlo Localization method is nothing but the repetition of *sensing* and *moving*.



- **Entropy**- a measure of information that the distribution has?



# Localization



- Localization
  - Basic Problem
  - Robot Movement
  - Robot Sense and Movement
  - **Summary**

## Summary

- Localization involves a robot constantly updating its *probability distribution* over the *sample space*. *Probability distribution* should have two qualities:
  - It should have one sharp peak. This indicates that the robot has a very specific belief about its current location.
  - The peak should be correct! If this weren't true, the robot would believe--very strongly--that it was in the wrong location.



Defining futures

# Localization



- Localization
  - Basic Problem
  - Robot Movement
  - Robot Sense and Movement
  - **Summary**

## Summary

- Steps of Monte Carlo localization can be written as:
  - Start with a belief set about our current location.
  - Multiply this distribution by the results of our *sense* measurement.
  - Normalize the resulting distribution.
  - Move by performing a *convolution*. This sounds more complicated than it really is: this step really involved multiplication and addition to obtain the new probabilities at each location.
- Keep in mind that step 2 always increased our knowledge and step 4 always decreased our knowledge about our location.



Defining futures

# Localization



- Localization
  - Basic Problem
  - Robot Movement
  - Robot Sense and Movement
  - **Summary**

## Summary

- *Mathematic Background*: Bayes Filter
- *Markov Localization*:
  - *Central Idea*: represent the robot's belief by a probability distribution over possible positions, and uses Bayes' rule/ convolution to update belief when robot senses or moves
  - *Markov Assumption*: past and future data are independent if one knows the current state
- *Kalman Filtering*
  - *Central Idea*: localization posed as sensor fusion problem
  - *Assumption*: gaussian distribution function
- *Particle Filtering*
  - *Central Idea*: Sample-based, nonparametric Filter
  - Monte-Carlo method
- *SLAM* (simultaneous localization and mapping)



Defining futures

# Localization



- Localization
  - Basic Problem
  - Robot Movement
  - Robot Sense and Movement
  - **Summary**

## Markov Localization

- Applying probability theory to robot localization
- Markov localization uses an *explicit, discrete representation for the probability of all position in the state space*.
- This is usually done by representing the environment by a *grid* or a *topological graph* with a *finite number of possible states* (positions).
- During each update, the *probability for each state* (element) of the *entire space* is updated.



Defining futures

# Localization



- **Localization**
  - Basic Problem
  - Robot Movement
  - Robot Sense and Movement
  - **Summary**

## **Probabilistic Robotics**

- Falls in between model-based and behavior-based techniques
  - There are models, and sensor measurements, but they are assumed to be incomplete and insufficient for control
  - Statistics provides the mathematical glue to integrate models and sensor measurements
- Basic Mathematics
  - Probabilities
  - Bayes rule
  - Bayes filters



*Defining futures*



# Probability



- Localization
- Probability
  - Axioms

## Axioms

$P(A)$  denotes probability that proposition  $A$  is true.

- $0 \leq P(A) \leq 1$
- $P(\text{True}) = 1$                        $P(\text{False}) = 0$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



Defining futures

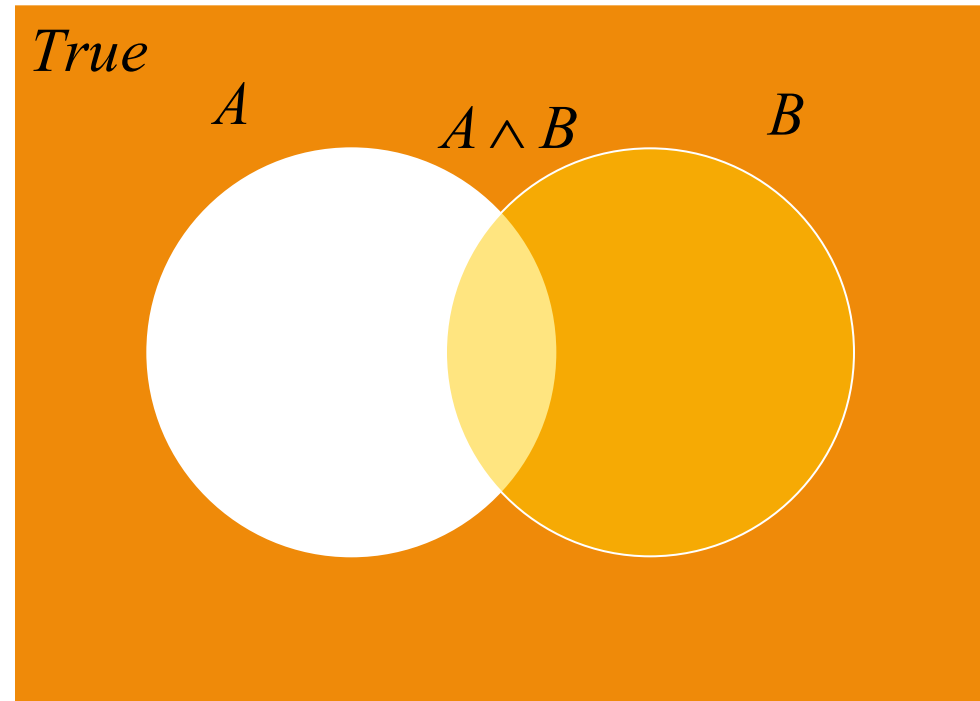
# Probability



- Localization
- Probability
  - Axioms

## Axioms

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Defining futures

# Probability



- Localization
- Probability
  - Axioms
  - Discrete Random Variables

## Discrete Random Variables

- $X$  denotes a *random variable*.
- $X$  can take on a countable number of values in  $\{x_1, x_2, \dots, x_n\}$ .
- $P(X = x_i)$ , or  $P(x_i)$ , is the *probability* that the random variable  $X$  takes on value  $x_i$ .
- $P()$  is called *probability density function*.
- E.g.

$$P(Room) = \langle 0.7, 0.2, 0.08, 0.02 \rangle$$



Defining futures

# Probability



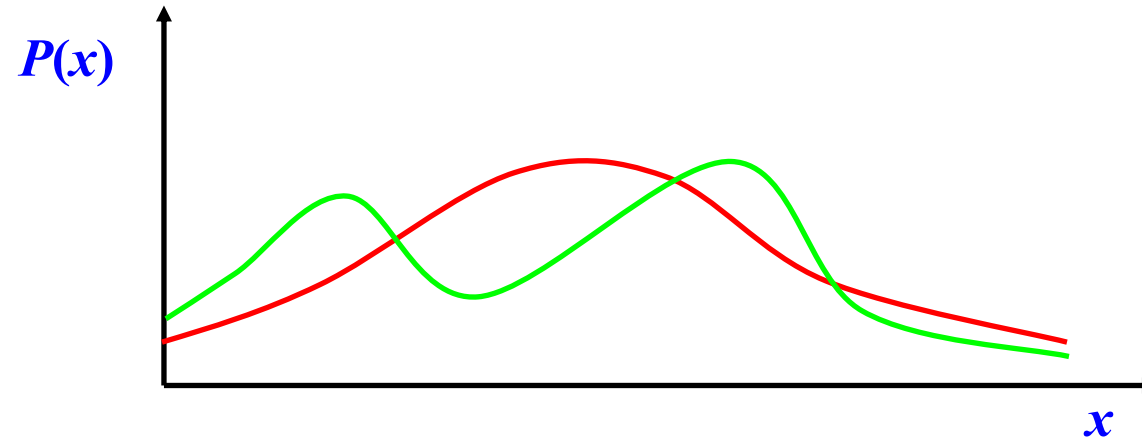
- Localization
- Probability
  - Axioms
  - Discrete Random Variables
  - **Continuous Random Variables**

## Continuous Random Variables

- $X$  takes on values in the continuum.
- $P(X = x)$ , or  $P(x)$ , is a *probability density function*.

$$P(x \in (a, b)) = \int_a^b p(x) dx$$

- E.g.



Defining futures

# Probability



- Localization
- **Probability**
  - Axioms
  - Discrete Random Variables
  - Continuous Random Variables
  - **Probability Theory**

## Probability Theory

- $P(X = x \text{ and } Y = y) = P(x, y)$
- If  $X$  and  $Y$  are *independent* then
$$P(x, y) = P(x) P(y)$$
- $P(x | y)$  is the probability of  $x$  given  $y$ 
$$P(x | y) = \frac{P(x, y)}{P(y)}$$
$$P(x, y) = P(x | y) P(y)$$
- If  $X$  and  $Y$  are *independent* then
$$P(x | y) = P(x)$$



Defining futures

# Probability



- Localization
- **Probability**
  - Axioms
  - Discrete Random Variables
  - Continuous Random Variables
  - **Probability Theory**

## Probability Theory

### *Discrete case*

$$\sum_x P(x) = 1$$

$$P(x) = \sum_y P(x, y)$$

$$P(x) = \sum_y P(x | y) P(y)$$

### *Continuous case*

$$\int p(x) dx = 1$$

$$p(x) = \int p(x, y) dy$$

$$p(x) = \int p(x | y) p(y) dy$$



Defining futures



# Bayesian Filtering

- Localization
- Probability
- Bayesian Filtering
  - Bayes Theorem

## Bayes Theorem

$$P(x, y) = P(x | y)P(y) = P(y | x)P(x)$$

$$\Rightarrow P(x | y) = \frac{P(y | x) P(x)}{P(y)} = \frac{\text{likelihood} \cdot \text{prior}}{\text{evidence}}$$

If  $y$  is a new sensor reading

$p(x)$  → Prior probability distribution

$p(x|y)$  → Posterior probability distribution

$p(y|x)$  → Generative model, characteristics of the sensor

$p(y)$  → Does not depend on  $x$



Defining futures

# Bayesian Filtering



- Localization
- Probability
- **Bayesian Filtering**
  - **Bayes Theorem**

## Bayes Theorem

$$P(H|D) = \frac{P(H) \times P(D|H)}{P(D)}$$

- Lets say that you are at the Daewoo Bus Stand in Rawalpindi and you see that he is wearing a mufti (white shirt, grey pants)
- Now say we know a lot of statistics about the area
- You want to be able to say from that the data you know what is the probability that the person in front of you is a cadet from EME College given that he is wearing a mufti dress
- In order to do this we need some data !!!



Defining futures



# Bayesian Filtering



- Localization
- Probability
- **Bayesian Filtering**
  - **Bayes Theorem**

## Bayes Theorem

$$P(H|D) = \frac{P(H) \times P(D|H)}{P(D)}$$

- $P(EME\_Cadet) = 0.60$
- $P(mufti | EME\_Cadet) = 0.90$
- $P(\sim EME\_Cadet) = 0.40$
- $P(mufti | \sim EME\_Cadet) = 0.20$
- $P(EME\_Cadet | Mufti) = ???$

$$P(EME\_Cadet | Mufti) = \frac{0.60 \times 0.90}{0.60 \times 0.90 + 0.4 \times 0.2} = 0.87$$



Defining futures

# Bayesian Filtering



- Localization
- Probability
- **Bayesian Filtering**
  - **Bayes Theorem**

## Bayes Theorem

$$P(H|D) = \frac{P(H) \times P(D|H)}{P(D)}$$

- $P(EME\_Cadet) = 0.10$
- $P(mufti | EME\_Cadet) = 0.90$
- $P(\sim EME\_Cadet) = 0.90$
- $P(mufti | \sim EME\_cadet) = 0.20$
- $P(EME\_Cadet | Mufti) = ???$

$$P(EME\_Cadet | Mufti) = \frac{0.10 \times 0.90}{0.10 \times 0.90 + 0.9 \times 0.2} = 0.33$$

- ***Best friend example !!!***

